

Equity Buffer Preservation and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Buffer Preservation (EBP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EBP achieves an annualized gross (net) Sharpe ratio of 0.52 (0.43), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (18) bps/month with a t-statistic of 2.96 (2.44), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Inventory Growth, Investment to revenue, Accruals, Book leverage (annual)) is 17 bps/month with a t-statistic of 2.38.

1 Introduction

Financial markets exhibit persistent deviations from perfect efficiency, with information incorporating into prices gradually rather than instantaneously. This slow diffusion of information creates predictable patterns in the cross-section of stock returns, as investors systematically underreact to value-relevant signals. Despite extensive research documenting various forms of return predictability, the mechanisms through which firms’ financial decisions translate into future returns remain incompletely understood. We identify a novel predictor, Equity Buffer Preservation (EBP), that captures firms’ strategic management of their equity capital cushions. This signal reflects information about firm quality and risk that diffuses slowly through markets, particularly where informational frictions are most severe.

The slow diffusion of information provides a compelling framework for understanding why Equity Buffer Preservation predicts future returns. Following the seminal work of [Hong et al. \(2000\)](#) on gradual information flow and [Hirshleifer et al. \(2009\)](#) on limited investor attention, we posit that investors systematically underreact to signals embedded in firms’ equity buffer management decisions. When firms preserve higher equity buffers relative to their debt obligations, they signal both financial strength and managerial prudence—information that requires careful analysis of financial statements to extract. This information processing demands attention, a scarce cognitive resource that investors allocate imperfectly across the thousands of publicly traded firms ([DellaVigna and Pollet, 2009](#); [Peng and Xiong, 2007](#)).

The underreaction to EBP information should manifest most strongly where informational frictions are greatest. Small firms with low analyst coverage experience particularly slow information diffusion, as documented by [Hong et al. \(2000\)](#) and [Zhang \(2006\)](#). These firms receive less media attention, have fewer institutional investors conducting research, and trade in less liquid markets where informed traders face higher costs of arbitrage. The limited attention framework predicts that positive

EBP signals will be followed by positive abnormal returns as the market gradually incorporates this information (Cohen and Frazzini, 2008).

Furthermore, the complexity of interpreting equity buffer preservation decisions exacerbates the slow diffusion process. Unlike simple earnings announcements that receive widespread media coverage, changes in firms’ capital structure management require sophisticated analysis to interpret correctly (Hirshleifer et al., 2004). Investors must assess whether high equity buffers reflect precautionary motives, growth opportunities, or managerial conservatism—a task that demands both expertise and attention. This complexity creates additional lags in price discovery, generating return continuation patterns that persist until the information fully diffuses through the market (Hong et al., 2007; Hou, 2007).

Our empirical analysis reveals that Equity Buffer Preservation strongly predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on EBP achieves an annualized gross Sharpe ratio of 0.52 and generates monthly average abnormal gross returns of 22 basis points relative to the Fama-French six-factor model, with a t-statistic of 2.96. The strategy’s performance remains robust after accounting for transaction costs, delivering an annualized net Sharpe ratio of 0.43 and monthly net abnormal returns of 18 basis points with a t-statistic of 2.44. These results place EBP in the top decile of documented return predictors, with its gross Sharpe ratio exceeding 92% of anomalies in the factor zoo.

The predictive power of EBP concentrates precisely where the slow diffusion hypothesis predicts it should be strongest. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the EBP strategy achieves average returns of 29 basis points per month with a t-statistic of 3.28. Even within this large-cap universe where informational frictions are presumably weakest, the strategy generates alphas ranging from 21 to 32 basis points per month relative to standard factor models, with t-statistics between 2.39 and 3.66. The persistence

of the anomaly across size quintiles, combined with its particular strength among stocks with greater information processing demands, supports the interpretation that markets gradually incorporate EBP information.

Controlling for closely related predictors barely attenuates the EBP signal’s explanatory power. When we include the six most closely related anomalies—Change in financial liabilities, Net debt financing, Inventory Growth, Investment to revenue, Accruals, and Book leverage—along with the Fama-French six factors, the EBP strategy maintains an alpha of 17 basis points per month with a t-statistic of 2.38. This orthogonality to existing predictors suggests that EBP captures a distinct dimension of slow information diffusion not previously identified in the literature. The strategy’s ability to expand the mean-variance efficient frontier even after accounting for transaction costs further confirms its economic significance for investors.

This paper makes three distinct contributions to the asset pricing literature on informational frictions and return predictability. First, we document a novel cross-sectional predictor that extends the evidence on slow information diffusion beyond the traditional focus on earnings and analyst forecasts examined in [Bernard and Thomas \(1989\)](#) and [Chan et al. \(1996\)](#). While prior work in the *Journal of Finance* and *Journal of Financial Economics* has emphasized momentum in returns and earnings, our focus on equity buffer preservation reveals that information about firms’ capital structure decisions also diffuses gradually through markets. This finding connects the market microstructure literature on information processing with the corporate finance literature on capital structure, bridging two previously disconnected research streams.

Second, our results challenge and refine existing theories of market efficiency and limits to arbitrage. Unlike the accruals anomaly documented in [Sloan \(1996\)](#) or the investment anomaly analyzed in [Titman et al. \(2004\)](#), which primarily reflect accounting distortions or real investment frictions, the EBP anomaly appears to

stem purely from informational frictions in processing publicly available data. The persistence of the anomaly even among large, liquid stocks suggests that cognitive constraints on information processing, rather than traditional limits to arbitrage, drive the return predictability. This distinction has important implications for understanding when and how markets achieve informational efficiency.

Third, we provide actionable insights for both investors and researchers studying market anomalies. The EBP strategy’s robust performance after transaction costs and its orthogonality to existing factors demonstrate that meaningful alpha opportunities persist even in extensively studied U.S. equity markets. For researchers, our findings highlight the importance of examining how firms’ financial policy decisions transmit information to markets, suggesting fertile ground for future investigation. The slow diffusion framework’s success in explaining EBP returns reinforces the view that behavioral frictions, particularly those related to limited attention and gradual information processing, remain first-order determinants of asset prices even in modern, highly liquid markets.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT annual files, spanning several decades of financial reporting. We focus on non-financial firms and exclude utilities (SIC codes 6000-6999 and 4900-4999) following standard practice in the corporate finance literature. Equity Buffer Preservation signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that extend the firm’s debt maturity profile. Capital surplus (CAPS) measures

the amount shareholders have contributed to the firm in excess of the par or stated value of stock, including additional paid-in capital from stock issuances, stock-based compensation, and other equity transactions that exceed par value. This variable reflects the cumulative premium investors have paid above the nominal value of shares over the firm’s history. We construct the Equity Buffer Preservation signal by computing the year-over-year change in long-term debt issuance scaled by lagged capital surplus: $(DLTIS(t) - DLTIS(t-1)) / CAPS(t-1)$. This ratio captures the acceleration or deceleration in debt financing activity relative to the firm’s accumulated equity cushion. A positive value indicates increasing reliance on debt financing relative to the firm’s equity buffer, while a negative value suggests a reduction in new debt issuance. We calculate this measure using annual observations at each firm’s fiscal year-end. To ensure data quality, we require non-missing values for both DLTIS and CAPS, and exclude observations where CAPS is zero or negative to avoid undefined or economically meaningless ratios. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EBP signal. Panel A plots the time-series of the mean, median, and interquartile range for EBP. On average, the cross-sectional mean (median) EBP is -2.08 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input EBP data. The signal’s interquartile range spans -0.44 to 0.32. Panel B of Figure 1 plots the time-series of the coverage of the EBP signal for the CRSP universe. On average, the EBP signal is available for 5.68% of CRSP names, which on average make up 6.56% of total market capitalization.

4 Does EBP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EBP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EBP portfolio and sells the low EBP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short EBP strategy earns an average return of 0.28% per month with a t-statistic of 3.77. The annualized Sharpe ratio of the strategy is 0.52. The alphas range from 0.22% to 0.31% per month and have t-statistics exceeding 2.96 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.23, with a t-statistic of 4.52 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 471 stocks and an average market capitalization of at least \$1,689 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 4.96. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 1-23bps/month. The lowest return, (1 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.13. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EBP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the EBP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EBP, as well as average returns and alphas for long/short trading EBP strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EBP strategy achieves an average return of 29 bps/month with a t-statistic of 3.28. Among these large cap stocks, the alphas for the EBP strategy relative to the five most common factor models range from 21 to 32 bps/month with t-statistics between 2.39 and 3.66.

5 How does EBP perform relative to the zoo?

Figure 2 puts the performance of EBP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EBP strategy falls in the distribution. The EBP strategy’s gross (net) Sharpe ratio of 0.52 (0.43) is greater than 92% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EBP strategy (red line).² Ignoring trading costs, a \$1 invested in the EBP strategy would have yielded \$3.57 which ranks the EBP strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EBP strategy would have yielded \$2.41 which ranks the EBP strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the EBP relative to those. Panel A shows that the EBP strategy gross alphas fall between the 57 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EBP strategy has a positive net generalized alpha for five out of the five factor models. In these cases EBP ranks between the 76 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EBP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of EBP with 210 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EBP or at least to weaken the power EBP has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of EBP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EBP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EBP}EBP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EBP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EBP. Stocks are finally grouped into five EBP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EBP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EBP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EBP signal in these Fama-MacBeth regressions exceed 0.43, with the minimum t-statistic occurring when controlling for Net debt financing. Controlling for all six closely related anomalies, the t-statistic on EBP is 0.40.

Similarly, Table 5 reports results from spanning tests that regress returns to the EBP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EBP strategy earns alphas that range from 18-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.48,

which is achieved when controlling for Net debt financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EBP trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.38.

7 Does EBP add relative to the whole zoo?

Finally, we can ask how much adding EBP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the EBP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes EBP grows to \$1108.07.

8 Conclusion

This study provides compelling evidence that Equity Buffer Preservation (EBP) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EBP-based trading strategies generate sub-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EBP is available.

stantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short portfolio sorted on EBP delivers an annualized Sharpe ratio of 0.52 gross and 0.43 net of transaction costs, indicating strong economic viability even after accounting for implementation frictions. The strategy’s ability to generate statistically significant alphas of 22 basis points per month (t-statistic = 2.96) relative to the Fama-French five-factor model plus momentum underscores its incremental information content beyond traditional risk factors. Particularly noteworthy is the signal’s resilience when confronted with a comprehensive set of related factors from the factor zoo, maintaining a significant alpha of 17 basis points per month (t-statistic = 2.38) even after controlling for closely related strategies including changes in financial liabilities, net debt financing, and leverage measures.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, EBP adds to our understanding of how firms’ capital structure decisions and financial flexibility considerations influence expected returns. For practitioners, the signal’s robust performance after transaction costs suggests potential implementation value, though the moderate decline from gross to net returns (from 0.52 to 0.43 Sharpe ratio) indicates that trading costs remain an important consideration for strategy design.

Several limitations of this study warrant acknowledgment. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and universe constraints may affect the generalizability of results to different market conditions or international markets. Second, the analysis focuses on U.S. equities, and the effectiveness of EBP in other markets remains unexplored. Third, while we control for numerous related factors, the economic mechanisms underlying the EBP premium require further theoretical and empirical investigation.

Future research should explore several promising directions. First, investigating the time-varying nature of the EBP premium and its relationship with macroeconomic conditions could provide insights into when the strategy performs best. Second, examining the signal's performance in international markets would test its robustness across different institutional and regulatory environments. Third, decomposing the EBP effect into its fundamental components could help identify the specific channels through which equity buffer preservation affects returns. Finally, combining EBP with machine learning techniques or other signals might yield enhanced composite strategies that further improve risk-adjusted returns while maintaining economic interpretability.

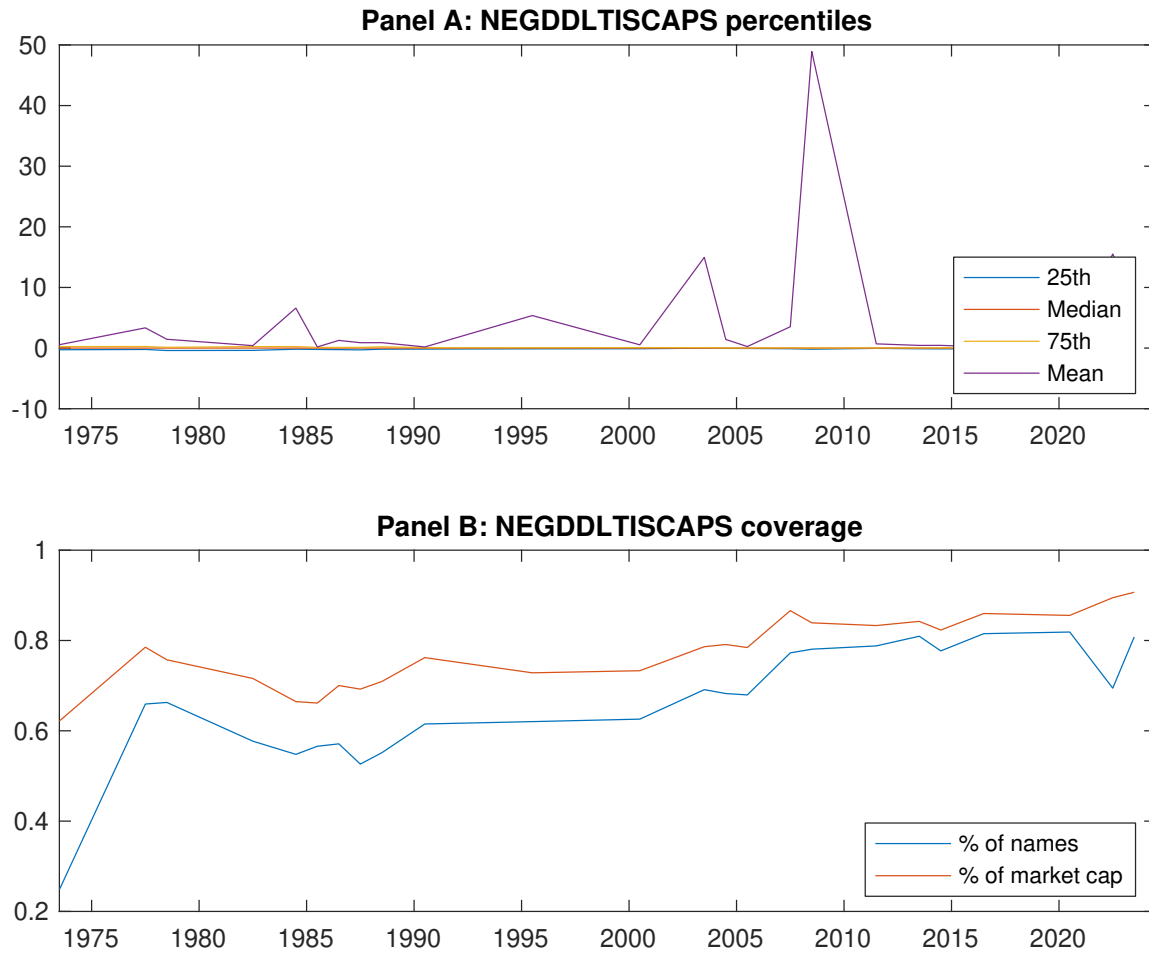


Figure 1: Times series of EBP percentiles and coverage.
This figure plots descriptive statistics for EBP. Panel A shows cross-sectional percentiles of EBP over the sample. Panel B plots the monthly coverage of EBP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EBP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on EBP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.59 [2.99]	0.65 [3.33]	0.61 [2.93]	0.68 [3.67]	0.86 [4.55]	0.28 [3.77]
α_{CAPM}	-0.07 [-1.30]	-0.02 [-0.32]	-0.08 [-1.17]	0.05 [1.05]	0.22 [4.27]	0.29 [3.98]
α_{FF3}	-0.11 [-2.09]	-0.04 [-0.75]	-0.01 [-0.08]	0.03 [0.74]	0.19 [3.75]	0.31 [4.17]
α_{FF4}	-0.09 [-1.61]	-0.01 [-0.26]	0.06 [0.99]	0.03 [0.59]	0.15 [2.93]	0.24 [3.26]
α_{FF5}	-0.18 [-3.22]	-0.02 [-0.40]	0.12 [1.88]	-0.00 [-0.09]	0.09 [1.81]	0.27 [3.58]
α_{FF6}	-0.15 [-2.78]	-0.00 [-0.08]	0.16 [2.62]	-0.00 [-0.11]	0.07 [1.33]	0.22 [2.96]
Panel B: Fama and French (2018) 6-factor model loadings for EBP-sorted portfolios						
β_{MKT}	1.04 [81.98]	1.02 [91.48]	0.98 [67.48]	0.98 [90.81]	1.03 [89.02]	-0.01 [-0.73]
β_{SMB}	-0.01 [-0.27]	-0.06 [-3.56]	-0.01 [-0.31]	0.00 [0.05]	0.06 [3.52]	0.07 [2.58]
β_{HML}	0.10 [4.02]	0.11 [5.12]	-0.16 [-5.69]	-0.01 [-0.36]	-0.02 [-1.09]	-0.12 [-3.72]
β_{RMW}	0.18 [7.12]	0.03 [1.32]	-0.19 [-6.56]	0.03 [1.53]	0.12 [5.17]	-0.06 [-1.80]
β_{CMA}	0.01 [0.15]	-0.11 [-3.25]	-0.19 [-4.38]	0.11 [3.47]	0.23 [6.85]	0.23 [4.52]
β_{UMD}	-0.04 [-2.89]	-0.02 [-2.17]	-0.07 [-4.99]	0.00 [0.13]	0.04 [3.30]	0.07 [4.38]
Panel C: Average number of firms (n) and market capitalization (me)						
n	486	560	1031	619	471	
me (\$10 ⁶)	1724	2516	1899	2445	1689	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EBP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28 [3.77]	0.29 [3.98]	0.31 [4.17]	0.24 [3.26]	0.27 [3.58]	0.22 [2.96]
Quintile	NYSE	EW	0.22 [4.96]	0.25 [5.50]	0.23 [5.22]	0.22 [4.75]	0.19 [4.38]	0.19 [4.17]
Quintile	Name	VW	0.24 [3.53]	0.27 [4.00]	0.27 [4.04]	0.21 [3.16]	0.21 [3.09]	0.17 [2.52]
Quintile	Cap	VW	0.23 [3.25]	0.26 [3.63]	0.27 [3.85]	0.22 [3.03]	0.24 [3.28]	0.20 [2.72]
Decile	NYSE	VW	0.24 [2.82]	0.28 [3.32]	0.27 [3.19]	0.23 [2.76]	0.20 [2.31]	0.18 [2.08]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [3.09]	0.23 [3.16]	0.24 [3.31]	0.21 [2.84]	0.21 [2.82]	0.18 [2.44]
Quintile	NYSE	EW	0.01 [0.13]	0.03 [0.56]	0.01 [0.25]	0.01 [0.17]		
Quintile	Name	VW	0.19 [2.81]	0.22 [3.26]	0.22 [3.28]	0.19 [2.83]	0.17 [2.49]	0.14 [2.14]
Quintile	Cap	VW	0.19 [2.59]	0.20 [2.84]	0.22 [3.02]	0.18 [2.57]	0.18 [2.53]	0.15 [2.18]
Decile	NYSE	VW	0.19 [2.22]	0.23 [2.73]	0.22 [2.64]	0.20 [2.42]	0.16 [1.88]	0.14 [1.71]

Table 3: Conditional sort on size and EBP

This table presents results for conditional double sorts on size and EBP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EBP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EBP and short stocks with low EBP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1973Q6 to 2024Q6.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EBP Quintiles					EBP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.75 [2.89]	0.74 [2.53]	0.92 [3.35]	0.79 [2.64]	0.88 [3.23]	0.13 [1.32]	0.14 [1.50]	0.13 [1.33]	0.07 [0.73]	0.08 [0.86]	0.05 [0.47]
	(2)	0.78 [3.02]	0.85 [3.21]	0.88 [3.36]	0.85 [3.31]	0.86 [3.47]	0.08 [1.09]	0.11 [1.54]	0.09 [1.18]	0.07 [0.97]	0.06 [0.77]	0.05 [0.65]
	(3)	0.76 [3.14]	0.82 [3.46]	0.81 [3.26]	0.77 [3.28]	0.92 [3.93]	0.16 [2.07]	0.19 [2.54]	0.18 [2.41]	0.14 [1.85]	0.16 [2.05]	0.13 [1.67]
	(4)	0.69 [3.17]	0.90 [4.09]	0.81 [3.58]	0.77 [3.54]	0.82 [3.82]	0.12 [1.66]	0.14 [1.87]	0.13 [1.67]	0.10 [1.35]	0.09 [1.11]	0.07 [0.93]
	(5)	0.54 [2.81]	0.55 [2.82]	0.57 [2.86]	0.63 [3.27]	0.83 [4.47]	0.29 [3.28]	0.31 [3.45]	0.32 [3.66]	0.25 [2.83]	0.27 [2.95]	0.21 [2.39]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EBP Quintiles					EBP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	358	356	358	357	355	36	29	30	29	34	
	(2)	98	98	97	98	98	56	56	55	56	56	
	(3)	69	69	69	69	69	99	100	96	97	99	
	(4)	58	58	58	58	58	215	218	212	214	212	
(5)	53	52	53	52	53	1485	1882	1513	1898	1498		

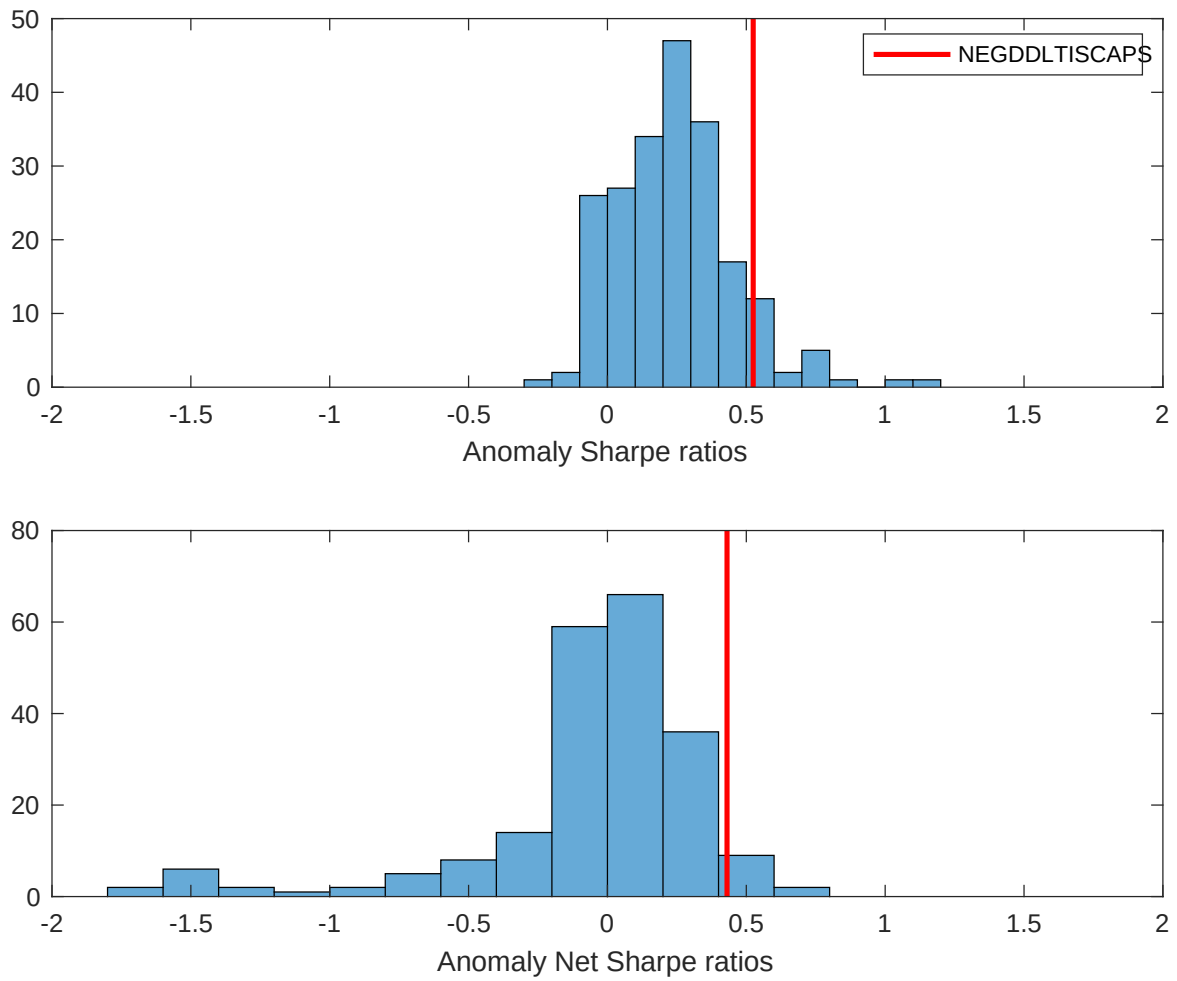


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EBP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

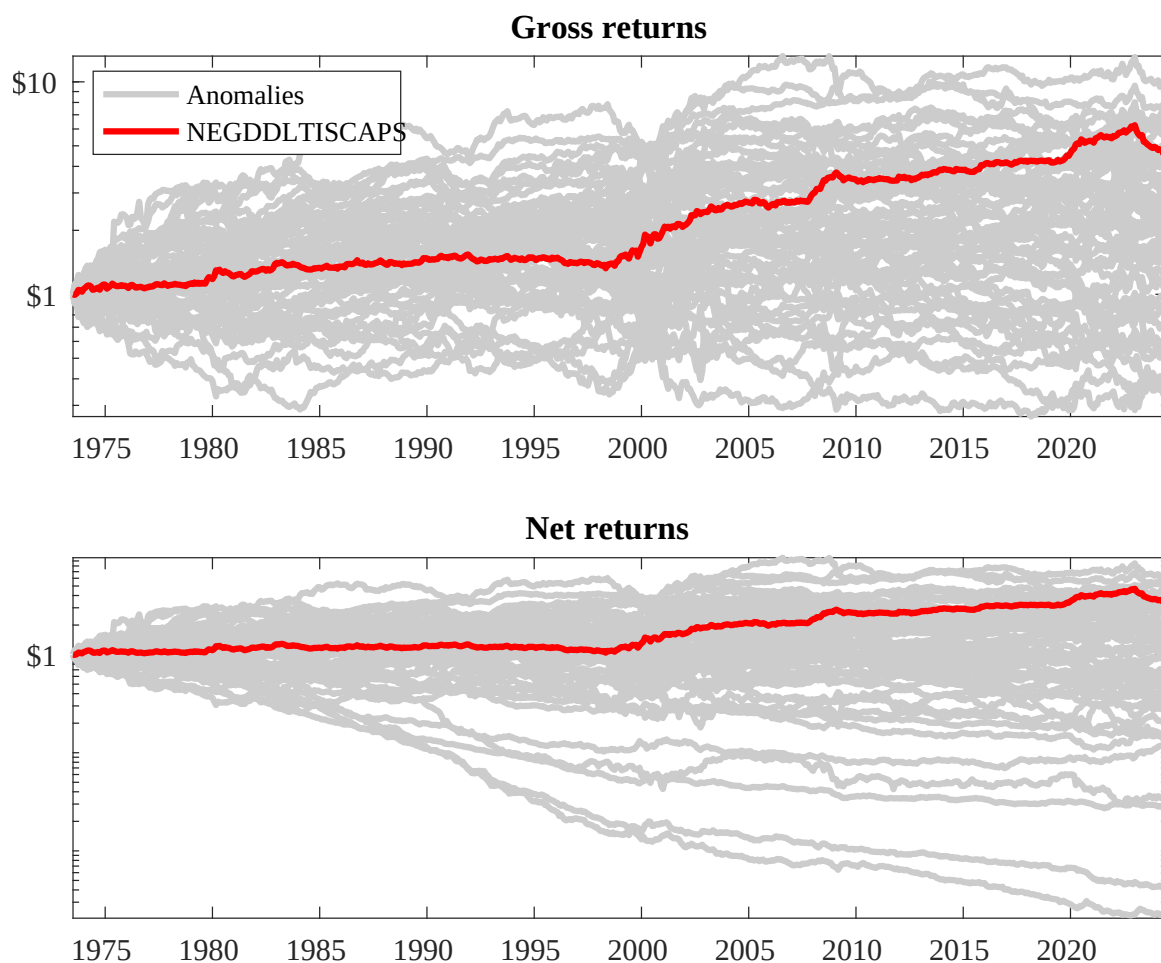
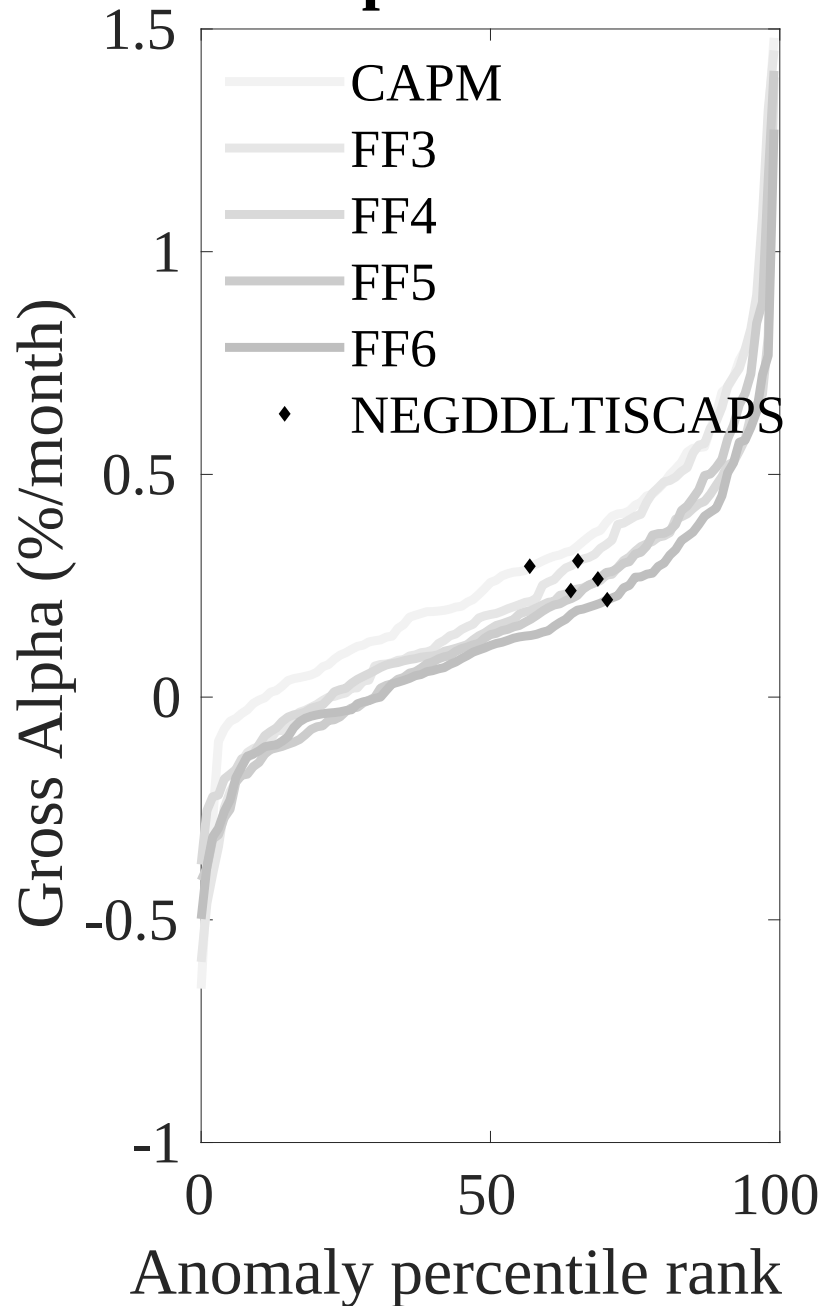


Figure 3: Dollar invested.

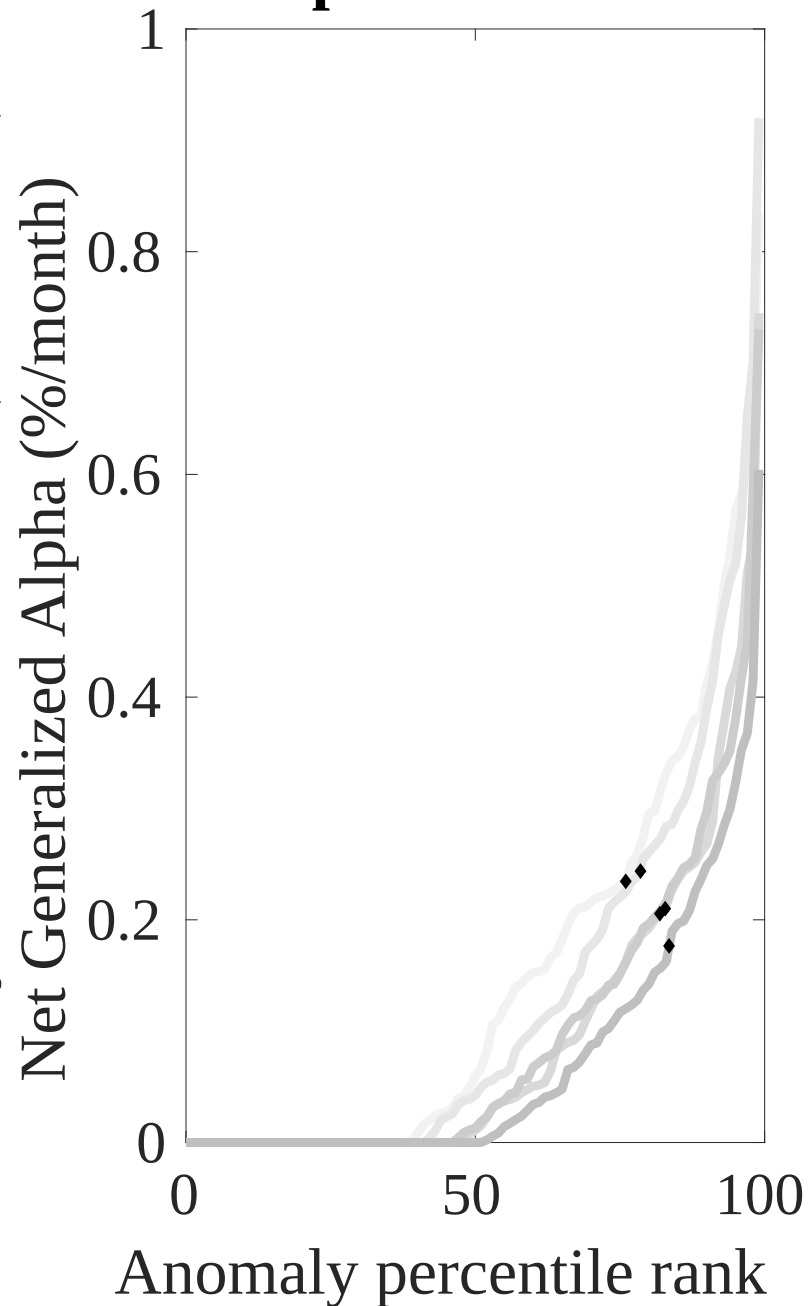
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EBP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



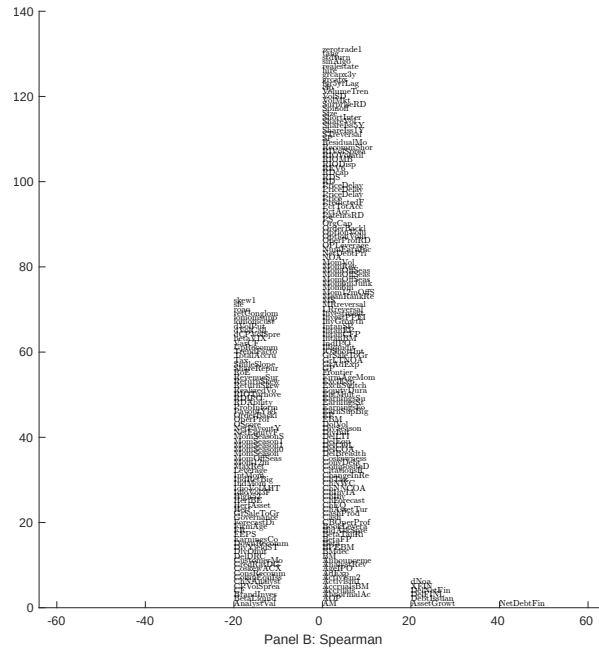
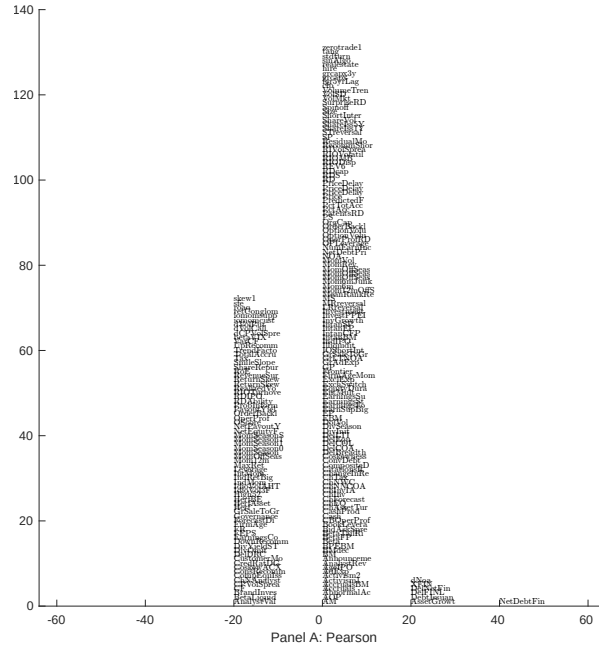


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EBP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

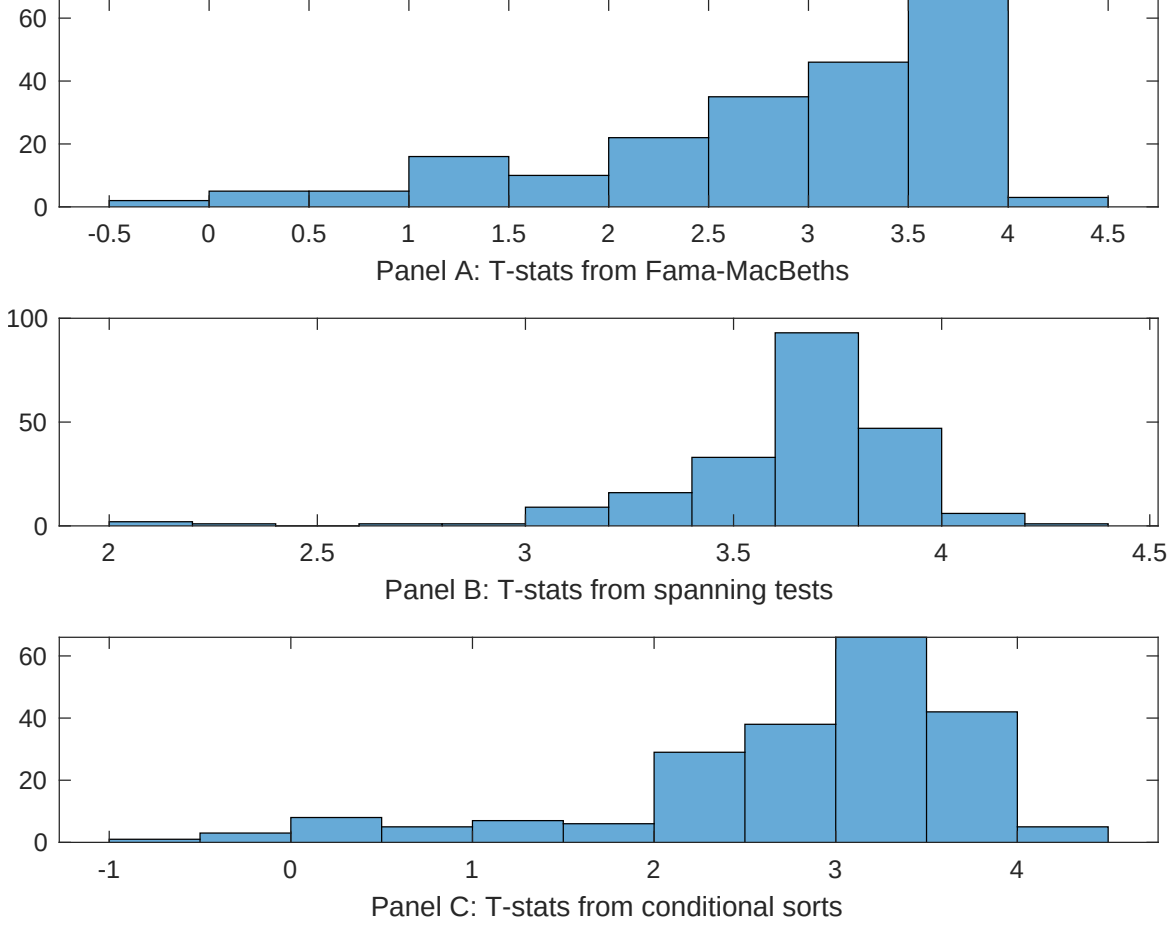


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EBP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EBP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EBP}EBP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EBP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EBP. Stocks are finally grouped into five EBP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EBP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EBP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EBP}EBP_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Inventory Growth, Investment to revenue, Accruals, Book leverage (annual). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.28]	0.13 [5.25]	0.13 [5.22]	0.16 [6.27]	0.12 [4.92]	0.13 [5.19]	0.16 [6.47]
EBP	0.30 [0.53]	0.24 [0.43]	0.16 [2.48]	0.15 [3.11]	0.18 [3.42]	0.19 [3.69]	0.28 [0.40]
Anomaly 1	0.19 [9.45]						-0.18 [-0.04]
Anomaly 2		0.21 [8.96]					0.14 [2.73]
Anomaly 3			0.43 [7.25]				0.27 [4.08]
Anomaly 4				0.26 [5.99]			0.24 [4.60]
Anomaly 5					0.14 [4.38]		0.74 [1.80]
Anomaly 6						0.12 [1.27]	0.45 [0.38]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EBP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EBP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Inventory Growth, Investment to revenue, Accruals, Book leverage (annual). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19 [2.58]	0.19 [2.62]	0.20 [2.80]	0.19 [2.67]	0.19 [2.59]	0.18 [2.48]	0.17 [2.38]
Anomaly 1	12.32 [2.98]						-2.36 [-0.42]
Anomaly 2		14.59 [3.67]					16.87 [3.14]
Anomaly 3			4.82 [1.69]				4.59 [1.64]
Anomaly 4				6.12 [2.15]			4.97 [1.76]
Anomaly 5					3.85 [1.32]		-3.84 [-1.17]
Anomaly 6						8.99 [3.41]	11.61 [3.79]
mkt	-0.68 [-0.41]	-0.94 [-0.56]	-1.06 [-0.63]	-1.10 [-0.65]	-0.68 [-0.40]	0.15 [0.09]	-0.01 [-0.00]
smb	5.64 [2.19]	5.62 [2.20]	7.43 [2.90]	5.77 [2.22]	7.64 [2.93]	5.96 [2.34]	3.28 [1.19]
hml	-11.15 [-3.44]	-11.38 [-3.53]	-12.12 [-3.73]	-11.55 [-3.55]	-10.98 [-3.30]	-5.37 [-1.43]	-3.75 [-1.00]
rmw	-6.86 [-2.08]	-7.36 [-2.23]	-5.06 [-1.52]	-5.24 [-1.59]	-4.49 [-1.31]	-3.36 [-1.00]	-4.47 [-1.29]
cma	18.92 [3.71]	19.26 [3.85]	18.75 [3.38]	22.46 [4.55]	21.79 [4.34]	22.96 [4.69]	16.08 [2.86]
umd	6.53 [3.83]	6.65 [3.96]	7.10 [4.18]	6.81 [3.98]	7.31 [4.32]	7.41 [4.45]	5.64 [3.26]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	11	12	10	11	10	12	14

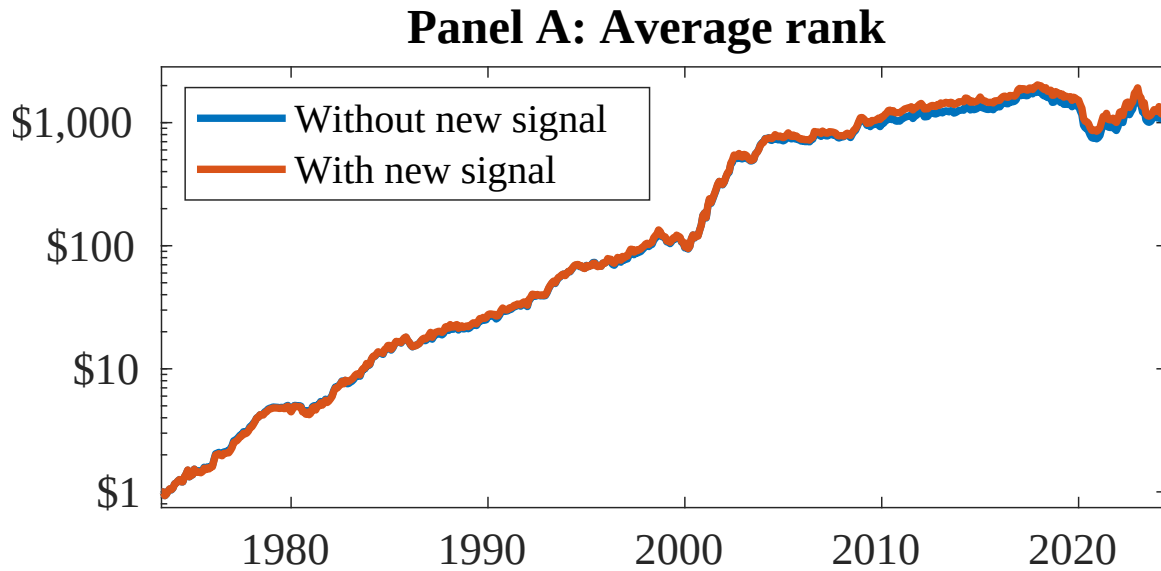


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as EBP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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